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REVIEWS

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AI-based smart glove for hand movement recognition and rehabilitation monitoring: a systematic review

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Abstract

Stroke-associated hand impairment is considered the major challenge in neurorehabilitation, which often causes loss of motor control and functional independence. While traditional therapeutic approaches have positive outcomes, they are labor-intensive and costly, and they lack the capacity for continuous performance monitoring. New technologies in wearable sensors and artificial intelligence (AI) have made it possible to produce smart gloves that can record detailed hand kinematics and, thereby, personalized therapeutic interventions. This is a systematic literature review of 101 peer-reviewed articles published between 2011 and 2025 that will address AI-driven smart glove systems incorporating flex sensors, inertial measurement units (IMUs), and force-sensitive resistors (FSRs) to identify gestures and gauge the rehabilitation condition. The review uses the PRISMA 2020 methodology to compare sensing modalities, architectures of algorithms, and clinical validation results. Results show that 72% of the studies used multimodal sensor fusion (flex + IMU). Recognition accuracy generally ranged between 82% and 99% across the included studies. The models based on deep learning, like convolutional neural networks (CNNs) and long short-term memory (LSTM) networks, demonstrate superior recognition performance, although they consume more computational resources; the classical models, like support vector machines (SVMs) and k-nearest neighbors (KNNs), are more computationally efficient and less resource-intensive. The remaining issues include sensor drift, insufficient diversity of the dataset, and large-scale clinical validation. The review emphasizes the necessity for future studies to concentrate on edge-optimized AI models, distinctive learning algorithms, and standardized clinical datasets, thus facilitating scalable, real-time therapy.

Keywords Smart gloves, Flex sensors, Machine learning, Rehabilitation, Hand gesture recognition

Introduction

Hand Gesture Recognition (HGR) is considered one of the key areas contributing significantly to the future of technology [1]. Neurological and musculoskeletal disorders such as stroke, rheumatoid arthritis (RA), and Parkinson's disease (PD) can have a significant

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effect on hand motor function [2, 3]. The technology using gloves for nonverbal communication has been continuously developed since the 1980s [4, 5]. Three common techniques of hand gesture recognition are vision, EMG (electromyography), and glove-based (smart gloves). Smart gloves have developed over the last forty years from the first "data gloves" to complex systems that incorporate textile-based sensors, force-sensitive resistors (FSRs), flex sensors, inertial measurement units (IMUs), and even optical fiber sensing technologies [6–9]. Compared to vision-based systems that are very sensitive to the lighting condition and occlusion, smart gloves offer constant and real-time functionality in a wide range of environments [10, 11]. In 1997, Humanglove, a spin-off of Scuola Superiore Sant'Anna, developed a highly precise joint-angle and thumb abduction/adduction device using 20 Hall-effect sensors (4 sensors per finger) [12, 13]. Though these innovations were made, early designs were costly, heavy, and lacked enough precision to be widely used in the clinic [14].

Between the 1990s and the 2000s, research transferred to assistive communication, developing gloves such as CyberGlove, which integrated flex sensors, accelerometers, and gyroscopes to translate sign language into text or speech. These developments expanded nonverbal communication possibilities but remained largely focused on static gestures. EMG-based systems often generated weak signals, while vision-based models suffered from environmental constraints. Various types of gloves have been designed: Fabric-based gloves, including HandTutor and Rapael Smart Glove, incorporate flex sensors or bend sensors and inertial measurement unit sensors (IMUs) to aid in motor recovery and physical therapy [15].

The development of smart gloves may be summed up into four generations (Fig. 1):

(1) early data gloves: virtual reality and motion-capture devices.
(2) assistive communication gloves for sign language translation;
(3) AI-driven gloves for static gesture recognition

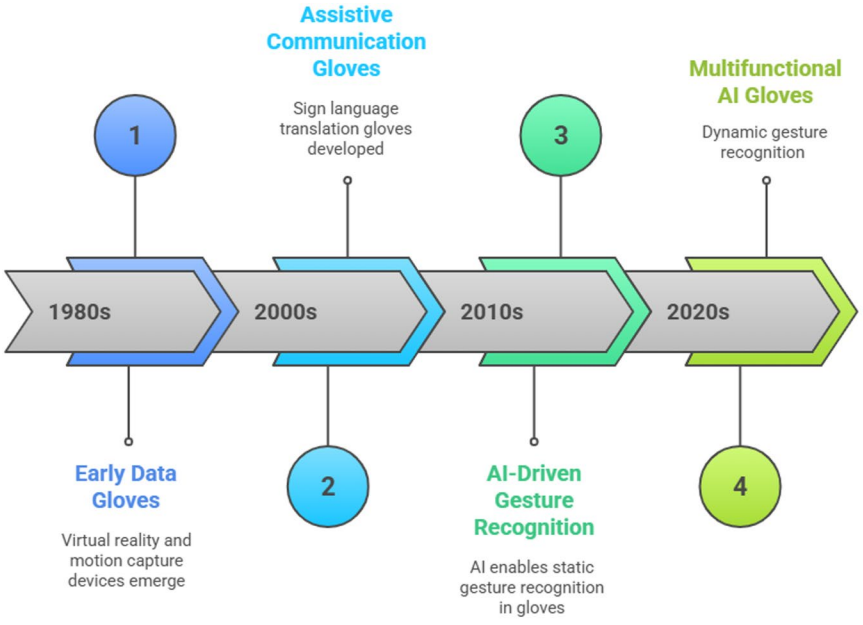


Fig. 1 Development of smart gloves

(4) multifunctional and AI-based gloves for dynamic gesture recognition and rehabilitation [16].

Nevertheless, there are still open challenges to balance accuracy, wearability, real-time AI processing, and personalization to promote rehabilitation. To direct the development of the upcoming generation of smart gloves and guarantee their progression from lab prototypes to extensively used assistive and healthcare technologies, these gaps must be filled [17, 18].

The combination of various sensing devices, including flex sensors that sense the bending of fingers, IMUs that read the orientation of the hand, and FSRs that link the force with which pressure is applied, has greatly enhanced the performance of smart gloves. By using machine learning methods, such as SVMs, Random Forests (RF), and CNNs, scientists have reached 90% accuracy in gesture recognition even in dynamic situations. The majority of systems are still only suitable for laboratory testing, with little clinical validation and scalability, even with significant improvements in sensor technology and AI-based gesture recognition.

Problem background

Stroke is one of the leading causes of disability and handicap worldwide, and in most countries, it is the second most common reason for death [19]. The most common cause of loss of upper extremity motor function, especially hand motor function, is stroke [20, 21]. Hand rehabilitation is a complex process that requires intensive approaches [22]. Cardiovascular disease often leads to impaired motor function in victims [23]. Spasticity and weakness are two major impairments that cause low quality of life and impaired overall recovery [24, 25]. Spasticity is an irregular and involuntary jerking movement due to hyperexcitability to the stretch reflex, while weakness is the crowding out of the inability to extend the muscles completely [26]. Post-stroke finger motor deficits are mainly caused by muscular weakness [27, 28]. Therefore, rehabilitation constitutes an inseparable part of post-stroke recovery [27]. Motor recovery requires neural plasticity, a biological process of re-forming neural pathways [29]. The process of neuroplasticity follows three steps. The initial phase entails cell death, which exposes new and secondary neurological networks, hence overturning cortical pathways of inhibition to excitatory pathways [30, 31]. Neural networks then restore connections through proliferation and later synaptogenesis in the neurons [32]. Much evidence proves that patients respond positively to neuroplasticity due to activity, exercise, and neural stimulation [33, 34]. The requirement for physical therapists is currently greater than the supply of skilled clinicians [13, 35]. Post-stroke hand impairment is commonly characterized by weakness, rigidity, and poor fine-motor control, which are major barriers to everyday activities. Even though repetitive, task-specific training is an effective method of facilitating neuroplastic recovery, traditional clinic-based treatment is still time-consuming and intermittent. At-home, remotely administered rehabilitation designed based on smart gloves incorporating flex, inertial, and force sensors provides quantitative and constant analysis of hand kinematics and grip dynamics. These wearable systems are easy to use and provide real-time feedback, making them a budget-friendly, data-driven option for long-term rehabilitation after a stroke.

Hand rehabilitation devices

Hand rehabilitation devices (HRDs) can be described as any equipment/brace used at any phase of the therapeutic management of the hand. HRDs are either orthoses, exoskeletons, or end-effector devices. Both forms provide the necessary value to the rehabilitation process. Orthoses are expected to offer needed assistance and act like the traditional hand braces. Exoskeletons and end-effector devices offer comprehensive rehabilitation to patients, combining various therapeutic remedies in one device, which are of various types as shown in Table 1. Actuation of rehabilitation devices improves motor recovery and overall recovery compared to regular in-person therapy with a physical therapist [36, 37]. Although no particular advantage or disadvantage has been established between exoskeletons and end-effector devices in regard to rehabilitation progress, the kinematic modes of joint flexion and extension are comparable [38, 39].

Methodology of research

Questions of research

The primary purpose of this research review is to determine the following:

Q1. What are the most common sensor technologies employed in smart gloves for hand gesture detection and rehabilitative control?

Q2: How is artificial intelligence used in smart glove systems to improve gesture recognition accuracy?

Q3. What are the limitations of current designs, and what gaps remain for future development?

Q4. How do various sensor combinations (flex, IMU, FSR, textile, inductive, and optical fiber) affect recognition performance and usability?

A schematic mapping of the formulated research questions is illustrated in Fig. 2.

All the research questions are provided in the Discussion section.

Selection criteria

This review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework to ensure methodological transparency and reproducibility. Title and abstract screening and full-text eligibility assessment were performed independently by two reviewers (AM and AMA); disagreements were resolved by consensus. This review was not pre-registered, and no protocol was prepared prior to conduct. A comprehensive literature search was undertaken across four major scientific databases—IEEE Xplore, ScienceDirect, SpringerLink, and MDPI—to identify relevant studies published between 2011 and 2025.

The search strategy employed targeted keywords, including *"smart glove," "flex sensor," "inertial measurement unit (IMU)," "gesture recognition," "artificial intelligence (AI),"* and *"rehabilitation."*

The following combined Boolean search string was used in all four databases: ("smart glove" OR "data glove") AND ("gesture recognition" OR "finger gesture") AND ("machine learning" OR "deep learning" OR "artificial intelligence") AND ("rehabilitation" OR "stroke" OR "hand motion") AND ("flex sensor" OR "SVM" OR "CNN" OR "LSTM").

For IEEE Xplore: ("smart glove" OR "data glove") AND ("gesture recognition" OR "finger gesture") AND ("machine learning" OR "deep learning" OR "artificial intelligence")

Table 1 Actuated devices and sensors with short descriptions

Name	Description
Hand MATE [40]	improved version of the HandSOME II device, including a linear actuator–driven, linkage-based system in place of non-motorized components. The metacarpophalangeal (MCP), proximal interphalangeal (PIP), and distal interphalangeal (DIP) joints may be actuated with precision and flexibility when force-sensing resistors (FSRs) are added, thus improving control and assistance during rehabilitation for hand mobility.
Tyromotion Amadeo [41]	The Amadeo by Tyromotion uses electromyographic (EMG) sensors and a linear rail mechanism to autonomously actuate each digit. There are significant improvements in patients' Fugl-Meyer Assessment (FMA) scores, indicating that it is effective in hand motor rehabilitation.
Gloreha [42]	The Gloreha is a pneumatically activated exoskeleton that will support the individual movement of each finger using a cable-driven system. This device has proven to improve the FMA scores of patients.
X-GLOVE [43]	cable-driven hand exoskeleton used for hand rehabilitation. It assists finger flexion and extension during passive rehabilitation exercises.
HEXORR [44]	HEXORR (Hand EXOskeleton Rehabilitation Robot) is an end-effector device that uses brushless DC motors and a linkage mechanism to assist finger flexion and extension.
Rutgers Master II [45]	The effectiveness of a modified Rutgers Masters II device was examined. an exoskeleton powered by air. The instrument uses tele-rehabilitation and EMG sensors to instantly improve FMA scores.
PowerAssist Glove [46]	Power Assist Gloves are pneumatic exoskeletons. Pressure sensors on each finger enable hand rehabilitation.
SEM Glove [47]	The SEM Glove is an exoskeletal device that is cable-driven. In combination with force-sensing resistors and linear actuators, it delivers passive rehabilitation regimens.
REHABGlove/ Hagshenas-Jaryani [48]	is an exoskeleton glove that uses pneumatic actuation. A combination of pressure and inertial sensors enables the device to administer rehabilitation.
Iron Hand [49]	The Iron Hand exoskeleton is a pneumatically actuated device. FSRs in this system help in rehabilitation interventions that produce low-level improvements in the strength of hands.
Hand of Hope [50]	Linkage-based linear actuators power the Hand of Hope. The device supports passive, active, and assisted rehabilitation exercises with EMG sensors.
WearMe [51]	WearMe is a cable-actuated exoskeleton device that is capable of rehabilitation. Utilizing inertial measurement units requires incorporation to ensure that rehabilitation is safe.
Exo Glove [52]	The linearly actuated Exo Glove exoskeleton system uses a linkage configuration to ensure safety. The device uses FSRs and EMG for safe rehabilitation.
Reha Digit [53]	The Reha-Digit is a new end-effector device that utilizes pneumatic cylinders to move the fingers about the MCP, PIP, and DIP joints. It is driven by DC motors and reports small improvements in FMA scores.
Power Grip [54]	The Power Grip exoskeletal framework provides secure actuation throughout the hand through a connection of a linkage-based framework combined with high-torque actuators.
Vanderbilt [55]	The Vanderbilt exoskeleton's brushless DC motors and link mechanism ensure safe hand actuation. Safety Hall-effect sensors accompany these motors.
Hand CARE [56]	The Hand CARE end effector is a cable-driven device that provides rehabilitation using brushless DC motors. Using FSRs, the device can control the rehabilitation precisely.
VAEDA [57]	VAEDA, a voice-activated exoskeleton, improves FMA scores slightly. We used brushed DC motors in a cable-driven architecture with electromyography and tension sensors to provide patients a safe and effective rehabilitation modality.
ReHand [58]	The ReHand exoskeleton is based on geared motors and a mechanism based on links to achieve a small increase in FMA scores. Using EMG sensors and voice control, the ReHand exoskeleton is capable of providing safe rehabilitation.
PneuGlove [59]	is a pneumatically powered glove, which proves the possibility of a small improvement of FMA scores. The device offers the safety of control by featuring force-bending sensors.
Tong et al. [60]	Tong et al. have developed an exoskeleton, which uses the linkage-based system and uses linear actuators to facilitate safe passive, active, and assistive rehabilitation. Using EMG sensors, this device has precise control over the hand.
Haptic Knob [61]	The end-effector device Haptic Knob provides moderate Fugl-Meyer Assessment (FMA) scores. This device uses brushed DC motors and pulleys and links to implement force sensors for precise movement.

Table 1 (continued)

Name	Description
Wege's device [62]	Wege et al.'s exoskeleton has brushless DC motors and a cable-operated system for safe passive rehabilitation. The device boosts patient strength with force and EMG sensors.
ExoK'ab [63]	The ExoK'ab is an exoskeleton device capable of accurate passive rehabilitation. Using force sensors in conjunction with the gear motors and gear trains, this device actuated each joint accurately.
VSFH [64]	is an exoskeleton with variable joint stiffness. Force-sensing resistors (FSRs) in a cable-driven system provide passive rehabilitation and support daily activities in the VSFH device.
Meeker et al. [65]	developed an exoskeleton device capable of passive rehabilitation. Meeker's system employs gearmotors and a cable-driven mechanism, utilizing EMG sensors to ensure safe actuation.
Flexo hand [66]	Flexohand is the only exoskeleton that can actuate each MCP, PIP, and DIP rotation independently. Passive rehabilitation consists of using cable-driven gear motors.
Delph et al. [67]	designed an exoskeleton for passive, active, and resistive rehabilitation. Actuate with servo motors and cables. Tension sensors and electromyography (EMG) detect user movement intentions and assist or resist. Adaptive control achieves therapeutic rehabilitation and safe, responsive actuation.

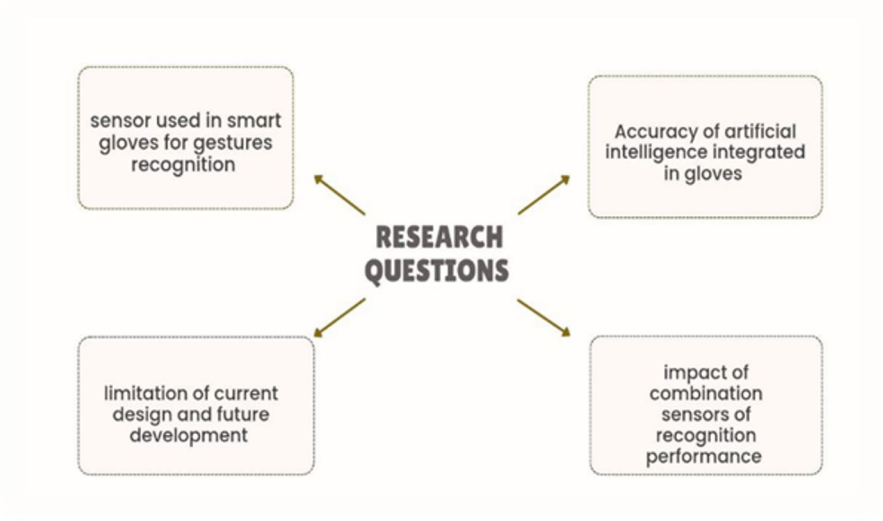


Fig. 2 Mapping the research questions

AND ("rehabilitation" OR "stroke") AND ("flex sensor" OR "SVM" OR "CNN" OR "LSTM").

For ScienceDirect: ("smart glove" OR "data glove") AND ("gesture recognition" OR "hand motion") AND ("machine learning" OR "deep learning") AND ("rehabilitation" OR "stroke"). For SpringerLink: ("smart glove" OR "wearable glove") AND ("gesture recognition" OR "hand gesture") AND ("artificial intelligence" OR "neural network") AND "rehabilitation." For MDPI: ("smart glove" OR "data glove") AND ("flex sensor" OR "IMU") AND ("machine learning" OR "CNN" OR "LSTM") AND ("rehabilitation" OR "stroke").

The selection process was structured in consecutive phases: an initial title and abstract screening, followed by a final full-text review, as illustrated in Fig. 3. A total of 765 articles were initially identified from the four databases, and 12 additional articles were identified from other sources, resulting in 777 articles. Of these, 185 were identified as duplicates and excluded, leaving 592 articles for screening. Of that total, 301 articles were discarded during the title and abstract screening process because they were not

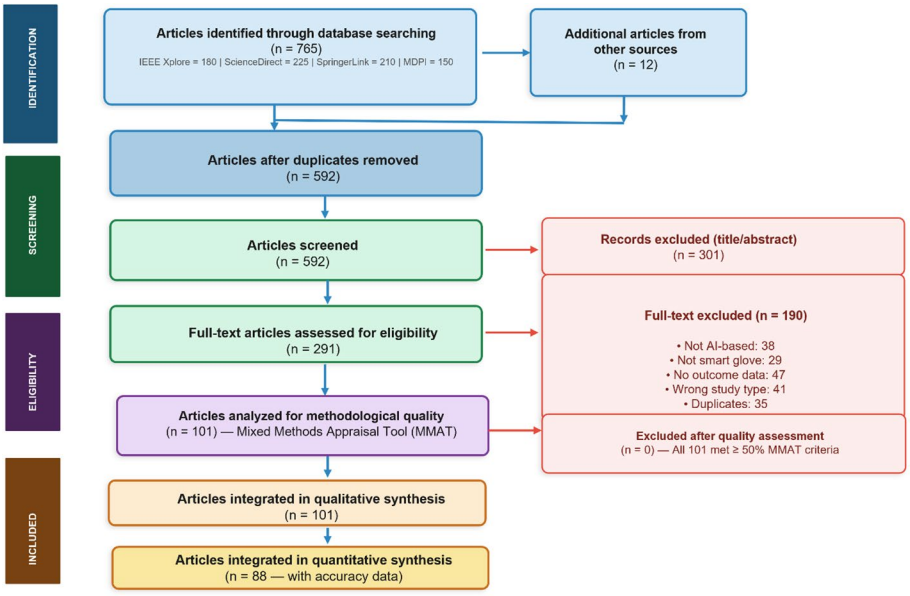


Fig. 3 PRISMA illustration of the articles' selection plan

related to wearable glove systems, hand gesture recognition, AI/ML methods, experimental validation, or rehabilitation monitoring. Consequently, 291 full-text articles were assessed for eligibility. During the full-text review phase, a further 190 articles were removed due to the following reasons: (i) not AI-based ($n = 38$); (ii) not a smart glove system ($n = 29$); (iii) no outcome data reported ($n = 47$); (iv) wrong study type ($n = 41$); and (v) duplicates identified at the full-text stage ($n = 35$). After methodological quality appraisal, 101 studies were included in the qualitative synthesis, while 88 articles with accuracy data were included in the quantitative synthesis.

The methodological quality of all included studies was evaluated with the help of a structured appraisal checklist based on the Mixed Methods Appraisal Tool (MMAT) and evaluating every study on four criteria: clarity of sensor configuration, description of the AI/ML methodology, reporting of experimental validation, and generalizability of findings. These studies were classified as high, moderate, or low quality, and only those studies of moderate or higher quality were subjected to comparative synthesis. Each study included was rated on the four MMAT-inspired criteria and given a study-level quality rating (high, moderate, or low); all of these ratings are provided in Supplementary Table 6 (attached at the end of this manuscript).

The distribution of the selected research by publishers and publication years is illustrated in Figs. 4 and 5.

Studies were deemed eligible if they reported on wearable smart-glove systems integrating one or more sensing modalities—such as flex sensors, force-sensitive resistors (FSRs), IMUs, electromyography (EMG), textile-based, or optical sensors—and utilized AI or machine learning (ML) algorithms, including Support Vector Machines (SVM), Random Forest (RF), Convolutional Neural Networks (CNN), or Long Short-Term Memory (LSTM). Additionally, only studies presenting experimental validation related to gesture recognition or rehabilitation monitoring were included. Conversely, conceptual studies without empirical testing and those based uniquely on vision-based recognition methods were excluded from the analysis. Table 2 shows the reference

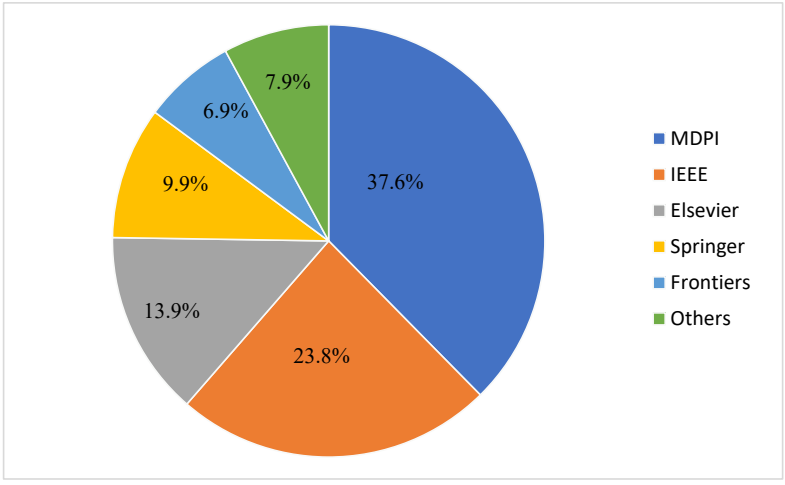


Fig. 4 Distribution of publishers

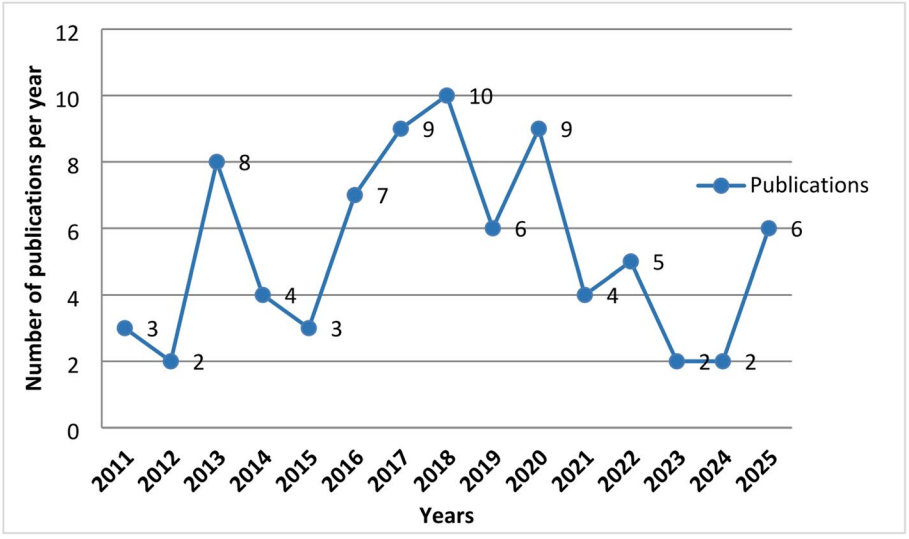


Fig. 5 Distribution of the selected papers by year

categorization of the reviewed studies. Note: Several references appearing in Table 2 predate the 2011–2025 primary inclusion window (e.g. [4]., 1980 [11], 1997 [10], 2000 [68], 2003). Such works are held only to provide historical background to the development of the smart glove technology and are not included in the systematic synthesis of the performance. This difference is shown by the updated inclusion criteria (Table 3).

The motivation for the study

The impairment of hand functioning is often caused by stroke and neuromuscular disorders, which, to a significant extent, affects the independence and quality of life of these patients. Although traditional rehabilitation approaches, despite their promise, are often time-intensive, dependent on continued supervision by a therapist, and only applicable in a clinical environment, limiting patient accessibility and continuity of therapy, these approaches are also beneficial. The improvements in wearable sensing devices and artificial intelligence (AI) have given promising opportunities for implementing continuous

Table 2 Reference categorization

Reference type	Author	Year	Reference type	Author	Year	Reference type	Author	Year
J	[1]	2019	J	[40]	2020	J	[69]	2020
J	[2]	2010	J	[41]	2018	J	[70]	2020
J	[3]	2010	J	[42]	2016	J	[71]	2020
J	[4]	2017	J	[43]	2015	J	[72]	2013
J	[5]	2011	C	[44]	2010	J	[73]	2016
J	[6]	2011	J	[45]	2006	C	[74]	2013
C	[7]	2022	J	[46]	2017	C	[75]	2013
J	[8]	2020	C	[47]	2012	J	[76]	2013
C	[9]	2018	J	[48]	2017	C	[77]	2014
J	[10]	2018	C	[49]	2019	C	[78]	2017
J	[11]	2003	J	[50]	2020	C	[79]	2009
J	[12]	2000	J	[51]	2019	J	[80]	2021
J	[13]	1997	J	[52]	2015	J	[81]	2022
J	[14]	2018	J	[53]	2008	J	[82]	2023
J	[15]	2020	J	[54]	2018	J	[83]	2025
J	[16]	2022	J	[55]	2017	C	[84]	2024
J	[17]	2024	J	[56]	2013	J	[85]	2025
J	[18]	2025	J	[57]	2016	J	[86]	2021
J	[19]	2008	J	[58]	2018	C	[87]	2022
J	[20]	2013	C	[59]	2010	J	[88]	2023
T	[21]	2007	J	[60]	2017	J	[89]	2025
J	[22]	2012	J	[61]	2007	C	[90]	2025
J	[23]	2018	J	[62]	2007	C	[91]	2025
J	[24]	2017	J	[63]	2016			
J	[25]	2013	J	[64]	2021			
J	[26]	1980	J	[65]	2017			
J	[27]	2009	J	[66]	2021			
J	[28]	2007	C	[67]	2014			
C	[29]	2022	J	[92]	2014			
J	[30]	2016	J	[93]	2015			
J	[31]	2014	J	[94]	2017			
J	[32]	2003	C	[95]	2020			
C	[33]	2019	J	[68]	2008			
J	[34]	2011	C	[96]	2003			
J	[35]	2016	J	[97]	2016			
C	[36]	2019	J	[98]	2018			
C	[37]	2018	J	[99]	2013			
J	[38]	2018	C	[100]	2019			
J	[39]	2020	J	[101]	2002			

C Conference, *J* Journal

Table 3 Inclusion and exclusion

Inclusions	Exclusions
Research on intelligent gloves Flex, FSR, IMU, EMG, textile, and optical fiber sensors.	Vision-based gesture systems without wearable sensors.
Studies investigating the integration of AI/ML techniques (e.g., SVM, CNN, and LSTM).	Works without data processing or machine learning models.
Experiments related to gesture recognition and rehabilitation monitoring.	Studies not focused on hand motion or rehabilitation.
Peer-reviewed journals, conferences, and papers from 2011 to 2025 (primary inclusion). Pre-2011 seminal works cited exclusively for historical context and not included in the systematic synthesis.	conceptual or theoretical works without experimental validation.

and home-based rehabilitation by creating smart gloves. The flex sensors and inertial measurement units (IMUs) in these gloves enable the precise monitoring of the movement of the wrist or fingers to produce statistical data of the condition of rehabilitation and motor rehabilitation. Nevertheless, numerous current prototypes are at the experimental phase and have not been well validated on clinical groups. Insufficient calibration, sensor drift, a lack of standardized rehabilitation datasets, and the restricted adaptability of AI models are some of the ongoing issues that prevent a practical application. This research was motivated by the need to produce accessible, flexible, and medically reliable smart rehabilitation gloves that integrate multimodal sensor integration with AI-driven motion analysis. The research examines the recent advances in the field with the aim of establishing the significant research gaps and proposing directions to be taken in the future to deliver scalable, personalized, and real-time rehabilitation solutions to individuals with motor deficits.

Inclusions and exclusions

A list of inclusion and exclusion criteria in this review is presented in Table 3.

Approaches to the analysis

One of the primary technologies of data gloves is the sensor that measures hand dexterity by detecting the bend, motion, rotation, and position of the hand [92]. Joint angle measurements are the most vital parameters of hand rehabilitation [93–95]. Recent studies on smart gloves with AI functionalities to identify gestures and rehabilitate the injured incorporate the dual concepts of analysis modeling and practical testing to provide trustworthy and interpretable results. Researchers are working to simulate the actions of several kinds of sensors, including flex, force-sensitive resistors (FSRs), inertial measurement units (IMUs), and electromyography (EMG) sensors, for different hand motions using computational models. The models enhance the accuracy of the systems as well as give meaningful observations to optimize sensor placement and configuration. The complementary experimental studies aim at creating and evaluating glove prototypes in reality, gathering motions, and using sophisticated machine learning models, such as support vector machines (SVM), convolutional neural networks (CNN), and long short-term memory (LSTM) networks. Combined theory and practice would result in thorough system assessment, enhanced flexibility, and clinical relevance in rehabilitation units.

Review of AI-based smart glove system

Sensor-based smart gloves for gesture recognition

Table 4 summarizes the major analysis approaches utilized in smart gloves, along with their advantages, disadvantages, and limitations. This method involves an excessive amount of sensors that are installed on a hand glove to sense bending, orientation, position, contact, and spacing of the fingers [68]. Flex sensor–based smart gloves have also been investigated for sign language and hand gesture recognition applications [102].

Various types of sensors are used on the glove, including flexible tubes with light, capacitive electrodes, flex sensors, and magnetic sensors to detect the curl of a finger. The key aim of the initial smart glove studies was to transform hand and finger motions into digital data by using embedded sensors. Tarchanidis and Lygouras introduced

Table 4 Advantages and disadvantages of approach of analysis

Approach of analysis	Advantage	Disadvantage	Limitation
Sensor-Based Approach (Flex, FSR, IMU)	• Low cost • Accurate • Uses a microcontroller such as ESP32 or Arduino • Simple	• Impact by noise and temperature • Deterioration	• frequent calibration. • Accuracy depends on glove fitting Inappropriate for rapid or multi-hand gestures.
AI-Based Approach (SVM, CNN, LSTM)	• High precision. • efficiently handling both static and dynamic motions. • Adaptable.	• large datasets for training • High computational cost • Overfitting with limited training data.	• Requires a microcontroller or external processor. • Real-time performance requires optimized lightweight models.
IoT-Based Rehabilitation Monitoring	• remote data transmission and tele-rehabilitation • real-time performance tracking and therapist feedback • Integrates with mobile apps and cloud platforms.	• Dependent on stable internet connectivity. • Higher power consumption • Security and privacy concerns in data handling.	• Limited Bluetooth/Wi-Fi range • Requires encryption for secure cloud communication.

a glove that incorporated flex sensors and a strain-gauge fiber of force to measure the applied force. A test was done by squeezing a ball, and the maximum force measured was about 85 N. Nevertheless, the system lacked an assessment of each of the fingertips, and it did not have assessments for palm force matching. Also, the fingers have to be placed on a rigid support, which will help make measurements more accurate. The flex sensors utilized had a large amount of error, making the force data inaccurate [96]. [103]. have developed a smart glove that can be used to convert sign language into speech output, which can consequently be heard, thus closing communication gaps that people with speech impairment face [97]. The prototype, which had been tested using Indian Sign Language, indicates promise for real-time sign language to speech transformation. This technology is used in robotics, gaming, and health care. Yudhana et al. used an MPU6050 inertial measurement unit (IMU) with five flex sensors to develop a data glove that monitors wrist orientation and finger flexing. The signals obtained were sorted through an Arduino Nano microcontroller and were used to recognize 26 alphabetic signs, which demonstrated stable and real-time translation of sign language applications [98]. Rajamohan incorporated five flex sensors with an added accelerator and a few tactile sensors in the glove. They could classify 13 gestures using these gloves [99]. In a related study, Chuang et al. designed a wireless smart glove system integrating five flex sensors with a microcontroller-type LilyPad Arduino to enable real-time and continuous hand gesture recognition. A gated recurrent unit (GRU) network and a maximum a posteriori (MAP) classification method were utilized to process the collection of sensor data. The experimental findings revealed an accuracy of recognition of 97.30%, which is higher than the performance of traditional deep-learning models, such as convolutional neural networks (CNNs) and long short-term memory (LSTM) models [100]. In a different experiment, Khan constructed a glove using five flex sensors per finger plus two additional sensors to measure hand tilt and rotation. Neural networks were used to classify, and the accuracy was 88.00% [101]. Lee and Bae developed a soft-sensor-based glove equipped with a sensor to monitor the motion of the metacarpophalangeal (MCP) and proximal interphalangeal (PIP) joints. The glove combined a hybrid system that included gesture-detection algorithms and simplification of motion sequences and LSTM-based recognition, thus obtaining an accuracy of 98.50% and a processing latency

of 12 ms per gesture. This contribution provided a basis in the development of robust sensor-based hand-gesture recognition and the use of a combination of sensing modalities, including flex sensors, force-sensitive resistors (FSRs), and IMUs in motion analysis to provide superior precision and adaptability [69].

Sensor topologies and mechanisms

There are promising opportunities for data gloves to assist in stroke patient rehabilitation measures and monitoring of other mobility diseases within the health care environment [70]. The evaluation of hand functionalities for clinical purposes requires the acquisition of multiple types of data, including, for example, pinching and gripping force, temperature, motion ranges of hand joints, and so on. Several typologies of sensors have been used to measure the physical quantities of interest, with circuit and architectural layouts specifically designed for it. The sensors generally used to monitor hand function belong to the following categories. All digits (little, ring, middle, index, and thumb) can perform the most outstanding action by bending to the palm (flexion) and then back (extension) to the original position. The popular sensor used in a data glove is a flexible sensor (resistant transducer). These types of flex sensors are intended to measure the degree to which a sensor is curved rather than straight and are widely embraced by researchers and engineers [71, 72, 93]. Flexible stretch-sensor-based gesture recognition is usually used to record changes in gesturing finger joints. Stretch sensors are often highly flexible, thinner, and more portable than other sensors [73]. The bending sensors normally used for this purpose belong to the following four categories: piezoelectric films [74], optical fibers [75], resistive transducers [76], and Hall-effect transducers [11]. Several strategies, described in the following, were used to evaluate hand position and orientation in space using inertial measurement units (IMU) and motion processing units (MPU), inclinometers and bending sensors [76], and inertial and magnetic measurement systems (IMMS) through Kalman filtering [77].

AI and machine learning-driven smart gloves

Machine learning is becoming common as it enhances the accuracy and speed of the system [78]. With advancements in artificial intelligence, machine learning algorithms began to play a key role in improving gesture classification and adaptability. Parvini used 22 flex sensors to measure hand bending, and the paper recommended use of a Multi-Layer Perceptron (MLP) and a Gesture Recognition algorithm using Bio-Mechanical Characteristics (GRUBC) to recognize American Sign Language (ASL). The maximum accuracy was 82.00% when GRUBC was applied to the 20 alphabet characters [79]. Panda et al. investigated various machine learning models, including SVM, KNN, logistic regression, and adversarial neural networks, using flex sensor data, where the adversarial model achieved the highest accuracy, 88.30% [80]. Lee et al. developed a smart glove that includes a support vector machine (SVM) classifier to identify objects with the help of five resistive flex sensors. The 91.90% recognition rate of the system among three separate objects of 160 training samples demonstrated the feasibility of the analysis using AI to interpret hand movement [16].

Faisal et al. have added flex sensors and an IMU to develop a multimodal deep learning glove based on a convolutional neural network (CNN) to fuse multimodal data. Both their models outperformed traditional classifiers in both static and dynamic gestures,

with F1 scores of 82.20% and 97.40% [81]. Ji et al. presented a 16 IMU, attention-based bi-directional long short-term memory (Attention-BiLSTM) network, and wearable glove, which is simple to wear and has a cost-effective feature, with a classification accuracy of 98.90% with Chinese sign language gestures [82]. More recently, Ravan et al. combined inductive sensors with Random Forest (RF), SVM, and KNN classifiers with the help of a generative adversarial network (GAN) used in data augmentation, achieving a 99.70% accuracy [83]. All these studies show how AI algorithms, especially CNN, LSTM, and SVM, can be used to provide powerful real-time recognition, noise cancellation, user versatility, and intelligent gesture recognition in wearable devices. Based on these, Wang et al. created a hybrid multi-feature neural network, which combined flex-sensor and IMU-sensor readings to boost the multimodal gesture recognition in rehabilitation. Their model had 99.00% classification accuracy, which was superior to the traditional CNN and SVM methods and highly flexible to various users [84]. Islam et al. developed a five-flex-sensor and a 9-axis inertia measurement unit (IMU) data glove based on an ESP32 microcontroller, which enabled real-time gesture detection using a hybrid classifier based on a random forest and an SVM. The data glove was found to be 98.30% accurate with minimal power consumption; therefore, it is applicable in continuous rehabilitation monitoring [85].

Smart gloves for rehabilitation and healthcare applications

Smart gloves have developed much more than their initial application in communication and gesture recognition as a required measure of rehabilitation and healthcare monitoring. Ahmed et al. designed a soft robotic glove fitted with flex sensors and servo motors that enables mirror-based rehabilitation therapy and achieves a higher level of motion coordination and mobility among paralyzed people [86]. Likewise, Ilyas et al. developed a low-priced rehabilitation glove, which combines flexion and force sensors to measure flexion, extension, and pinch strength among stroke survivors. The device reported statistically significant improvements in hand range of motion in a 15-day clinical trial, thus proving its value as an effective rehabilitation tool [87]. It is based on these developments that Imtiaz et al. presented a multifunctional glove that employs flex sensors, force-sensitive resistors (FSRs), and accelerators combined with an ESP32 microcontroller, which allows obtaining real-time data and monitoring the treatment at home. The device focuses on the low cost and familiarity of use [88]. Dhiman et al. added to this design by incorporating Internet of Things (IoT) devices, namely the MAX30100 biosensor and an OLED display set into a glove based on the ESP32 platform, allowing constant tracking of grip strength and heart rate as well as oxygen saturation with an error margin of ± 2 [89]. In addition, Hosie et al. proposed an open-palm style of gloves with the integration of five resistive flex sensors and a single inertial measurement unit (IMU). The system was found to have a high calibration accuracy with an error of the mean of less than 5 degrees, which allowed it to be of high comfort and long-term wearability due to its lightweight, ergonomic design [90]. The Flexible Inertial Poser (FIP) is an upcoming motion-capture device proposed by [104] that is aimed at combining flex and inertial sensors into typical clothing, thus making it possible to track the entire body of motion in the most precise and comfortable way. Two flex sensors utilizing four IMUs into fitted clothes, flex sensor examination by using a physics-informed calibrator, removing drift of the sensor by applying the Hidden Markov Model (HMM)-based drift correction,

and integrating multi-model measurement with a pose fusion predictor. The validation exhibited a 19.5% angular accuracy improvement, a 26.4% decline in the angular error of the elbow, and a 30.1% reduction in the positional error in comparison to the current state-of-the-art approaches and indicated the strong potential of the system in the applications of rehabilitation, athletic performance analysis, and the metaverse [91]. Combining machine learning, sensor fusion, and Internet of Things connectivity, today's smart gloves provide real-time performance monitoring, adaptive therapeutic feedback, and distance rehabilitation, thus providing new opportunities for the improvement of motor recovery and autonomous life in patients with a neurological or motor loss.

Challenges to be addressed

A critical limitation of the current literature is the absence of standardized comparative benchmarking. Across the reviewed studies, training set sizes ranged from fewer than 50 to over 2,000 samples; the number of classified gestures varied from 3 to 52; and validation strategies ranged from simple hold-out splits to leave-one-subject-out cross-validation. This heterogeneity makes direct accuracy comparisons highly misleading. Notably, several studies reporting accuracy above 95% used subject-dependent validation on small datasets ($n < 10$), raising the risk of overfitting. Future reviews and original studies should report the following:

(i) sample size and population composition (healthy vs. patient);
(ii) number and type of gestures;
(iii) validation strategy (cross-validation, hold-out, leave-one-subject-out);
(iv) whether clinical populations were included. Without these disclosures, pooled accuracy statistics should be interpreted with caution. Despite the considerable progress in the field of smart gloves to identify gestures and conduct rehabilitation monitoring, there are several essential issues to overcome before making them accessible to a large-scale clinical translation and commercial implementation. Firstly, sensor drift and signal degradation over time are significant constraints that have not been eliminated yet, especially with resistive flex sensors and force-sensitive resistor (FSR) sensors used throughout the higher rehabilitation periods. These impacts challenge the reliability and repeatability of data; thus, a key research focus is the creation of self-calibrating, drift-compensated, and adaptive sensing functions. Second, inter-subject variation caused by structural, motion, and physiological differences in hands limits the scope of generalization of artificial intelligence (AI) models. For this weakness to be corrected, it is necessary to develop large-scale, standardized, and publicly available multimodal datasets with cross-user validation procedures to increase robustness, fairness, and reproducibility. Third, energy consumption and efficient computational costs are associated with considerable limitations of real-time embedded AI. To obtain a trade-off between the accuracy of inference and energy consumption, lightweight and quantized neural network models should be optimized to work on low-power microcontrollers (e.g., ESP32, STM32, ARM Cortex-M). Fourth, usability in the long term is restricted due to the difficulty of ergonomic construction, comfort, and material durability. In the future, breathable, biocompatible, and washable materials should be introduced in the design of the sensors to maintain their flexibility and mechanical stability during repetitive use. Fifth, regulatory approval and clinical

adoption remain obstructed by the absence of large-scale clinical validation and interdisciplinary collaboration. Clinical translation of smart glove systems requires more than high classification accuracy in controlled settings. Future work must incorporate standardized rehabilitation outcome measures such as the Fugl-Meyer Assessment (FMA), the Action Research Arm Test (ARAT), and the Box and Blocks Test (BBT) to allow cross-study comparison and clinical benchmarking Prospective randomized controlled trials (RCTs) with adequate sample sizes and diverse patient populations are essential. From a regulatory standpoint, devices intended for therapeutic use in stroke rehabilitation must navigate frameworks such as the U.S. FDA 510(k) clearance pathway (as Class II medical devices) or the EU Medical Device Regulation (MDR 2017/745) CE marking process. Developers must address software as a medical device (SaMD) classification, cybersecurity requirements, and post-market surveillance obligations. Long-term patient adherence is also a critical but underexplored concern: current prototypes are rarely evaluated for usability over periods exceeding a few weeks, and factors such as glove discomfort, calibration burden, battery life, and ease of donning/doffing all affect real-world adoption. To transform lab prototypes into approved medical-grade rehabilitation mechanisms, comprehensive and statistically controlled studies are essential. Finally, the increased dependency on the Internet of Things (IoT) connectivity presents cybersecurity and privacy data threats that need to be addressed through the use of secure communication standards and data anonymization techniques. In conclusion, to implement the potential of smart glove technologies, it is necessary to have a multidisciplinary model that integrates materials science, embedded artificial intelligence, biomechanics, and clinical engineering. The combination of self-calibration sensor fusion, edge-optimization AI algorithms, and standard rehabilitation protocols will be essential in developing smart gloves into scalable, power-efficient, and clinically reliable rehabilitation systems capable of complete healthcare application. One important but overlooked aspect of smart glove development relates to the transition from research prototypes to medical devices. The Challenges section above outlines regulatory approval (FDA 510(k) and EU MDR 2017/745 CE marking), required clinical outcomes (FMA, ARAT, and BBT), and patient compliance. In short, none of the above studies have included these measures in a systematic clinical trial, highlighting the gap between the prototype design and the clinical validation of the systems. Future research is needed to account for human factors and patient-reported outcomes to enable home-based use.

Discussion

This section discusses and interprets the reviewed literature in relation to the research questions.

Q1. What are the most common sensor technologies employed in smart gloves for hand gesture detection and rehabilitative control?

Smart glove systems utilize several types of sensors to precisely monitor various hand motions of humans. Flex sensors and inertial measurement units (IMUs) are among the most widely used modalities, and they measure flexion of joints by measuring changes in electrical resistance or capacitance and dynamic motion parameters, respectively:

acceleration, angular velocity, and orientation. Also, force-sensitive resistors (FSRs) will be incorporated to measure grip force and pressure; they are essential when measuring rehabilitation progress. Utilizing textile-based, inductive, or optical fiber sensors, improved designs enhance wearability, flexibility, and the precision of measurements. These systems combine the data of various modalities to enable full kinematic analysis of finger and wrist motions, hence providing quantitative information that can be used in post-stroke rehabilitation.

Q2. How is artificial intelligence used in smart glove systems to improve gesture recognition accuracy?

Artificial intelligence (AI) is acknowledged as a critical component to enhance the accuracy, autonomy, and overall usability of smart glove systems utilizing gesture recognition. Mostly, two major computational models are used: deep learning and supervised machine learning. In supervised learning, Support Vector Machines (SVM), K-Nearest Neighbors (k-NN), and Random Forests (RF) algorithms typically can be trained to have recognition accuracies of between 85 and 92% on fixed gesture recognition, therefore giving efficient generalization to dynamic movement conditions that are limited. Deep learning systems like Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Generative Adversarial Networks (GANs), and Multi-Layer Perceptron Feedforward Neural Networks (MLP-FFNNs) can achieve 95 to 99% accuracy for complex time-based gestures. In addition, GAN-based data augmentation is applied to reduce the effects of data variations in small datasets and to optimize algorithm performance. In addition, the models can be implemented on embedded devices such as the ESP32 or Raspberry Pi and thus can provide real-time inference for telerehabilitation in portable, energy-efficient systems, as more recent studies have demonstrated. The methodological differences presented by the reviewed studies make it difficult to draw comparisons of performance. First, each study employed different validation schemes: single subject-based leave-one-out cross-validation (LOOCV) or k-fold cross-validation (kFCV) has been adopted in some studies, while data has been split into train and test sets in others, and the data has been subject-dependent. Few studies that perform leave-one-subject-out (LOSO) cross-validation, which better reflects the performance on new users, exist. Second, the number of subjects ranged from fewer than 5 to more than 30 (most studies recruited healthy individuals rather than post-stroke patients, who are the main target for clinical applications). Third, the number of gesture classes ranged from 5 to more than 50, with accuracy metrics needing to be standardized to be comparable. High accuracy (over 95%) reported in studies with small samples (less than 10 subjects), subject-dependent validation, and a small gesture set should be viewed with caution, as it could be heavily influenced by overfitting to the motion patterns of a single user. Future studies need to report on sample size and population (healthy or clinical), gesture set size, validation strategy, and data availability as detailed in Table 4 of this review. The minimum criteria for "clinical readiness" are testing with other datasets and prospective testing with new clinical populations. Table 5A representative subset of studies was selected from the 101 studies included in this review, using the following criteria:

(i) reporting of the number and type of participants (healthy; clinical).
(ii) clear validation strategy.

Table 5 Comparative overview of AI-based smart glove studies

Study (Year)	Participant	Population	No. of Gestures	Validation	Accuracy (%)	Clinical Validation
Chuang et al. (2019) [100]	10 subjects	Healthy	26	Hold-out split	97.30%	No
Lee & Bae (2020) [69]	8 subjects	Healthy	10	Subject-dependent split	98.50%	No
Faisal et al. (2022) [81]	15 subjects	Healthy	20 (static + dynamic)	k-fold CV	97.40% (F1)	No
Ji et al. (2023) [82]	20 subjects	Healthy	52 (Chinese SL)	LOSO cross-validation	98.90%	No
Ravan et al. (2025) [83]	12 subjects	Healthy	10	k-fold CV + GAN augment.	99.70%*	No
Wang et al. (2024) [84]	18 subjects	Healthy + rehab.	15	k-fold CV	99.0%	Partial (pilot)
Ilyas et al. (2022) [87]	20 stroke patients	Stroke patients	ROM monitoring	Pre/post clinical trial	N/A (rehab. outcome)	Yes (15-day)

* High accuracy (> 95%) on small data sets (< 10 subjects) and subject-based training should be considered with caution due to the risk of overfitting. CV = cross-validation; LOSO = leave-one-subject-out; SL = sign language; ROM = range of motion; RCT = randomized controlled trial

(iii) relevance to at least one of the four research questions, with a preference for studies with clinical populations or those reporting on rehabilitation outcomes.

Q3. What are the limitations of current designs, and what gaps remain for future development?

The prototype versions of smart gloves are largely limited. The main limitations involve sensor drift, individual calibration requirements, and biased data depending on the user. The absence of standardized multimodal data prevents the correct comparative analysis of algorithms. Additionally, wearability and power efficiency restrictions decrease the wearer's comfort and limit extended usage. There is only a small proportion of these systems that have been subjected to large-scale trials or have been medically certified and thus have not been translated into practice. As a result, subsequent versions must include self-calibration sensors, flexible artificial intelligence, and learning systems to obtain scalability, inter-patient flexibility, and reliable clinical implementation.

Q4. How do various sensor combinations (flex, IMU, FSR, textile, inductive, and optical fiber) impact recognition performance and usability?

Smart gloves with flex sensor and IMU integration are the most used, as they provide the balance between the cost and the accuracy. Glove performance and stability are dependent on the type and number of sensors adopted. With rehabilitating tracking, FSRs provide the ability to determine functional evaluation and measure gripping force. Although inductive and optical fiber sensors are more accurate, their production is more complex, costly, and fragile, leading to their unsuitability for complete rehabilitation. On the other hand, textile and stretch sensors enhance comfort and long-term wearability for the user but are affected by the unstable signal from environmental factors. In general, multimodal sensor fusion is more accurate and robust because it adjusts the weaknesses of specific sensor types, and it is the most promising in the case of adaptive smart glove systems in the future.

Conclusion

The smart glove technologies have progressed from the simple motion capture sensors to multimodal and intelligent technologies that interpret single gestures and monitor rehabilitation in real-time. The introduction of AI algorithms in conjunction with flex sensors and IMUs has greatly improved the accuracy of motion recognition; some studies report a greater degree of recognition of more than 95%. The results indicate that all of post-stroke rehabilitation, tele-rehabilitation, and continuous motor performance measurement are feasible applications of advanced information from the smart glove. There remain, however, major obstacles, ranging from sensor drift and low inter-subject generalizability to the lack of large-scale clinical trials and lack of clarity in the pathway towards commercialization and regulatory approval This requires three advances: First, AI models must be validated in both subject independent and population-representative settings in the presence of a known rehabilitation tool, such as the Fugl-Meyer Assessment (FMA) or the Action Research Arm Test (ARAT). Second, edge-optimized and multimodal datasets and self-calibrating sensor fusion architectures represent the primary research priorities for enabling long-term, real-time rehabilitative applications. Ultimately, the next generation of smart gloves, if validated in adequately powered clinical trials, has the potential to improve patient outcomes through clinical stability, comfort, durability, and regulatory compliance—enabling a transition from traditional therapy to personalized, data-driven rehabilitation (Table 6).

Table 6 Studies included ($n = 24$, all MMAT scores Moderate or High quality). Complete quality ratings are included for all 101 studies

Study (Year)	Sensor Type	AI/ML Algorithm	C1	C2	C3	C4	Rating	In Synthesis
Chuang et al. (2019) [100]	Flex + Arduino	Gate network classifier	Y	Y	Y	Y	H	Yes
Lee & Bae (2020) [69]	Flex sensors	SVM	Y	Y	Y	Y	H	Yes
Faisal et al. (2022) [81]	Flex + IMU	CNN (multimodal)	Y	Y	Y	Y	H	Yes
Ji et al. (2023) [82]	16 IMU	Attention-BiLSTM	Y	Y	Y	Y	H	Yes
Ravan et al. (2025) [83]	Inductive sensors	RF/SVM/KNN + GAN	Y	Y	Y	N	M	Yes
Wang et al. (2024) [84]	Flex + IMU	Hybrid neural network	Y	Y	Y	Y	H	Yes
Ilyas et al. (2022) [87]	Flex + force sensors	Rehab. Monitoring	Y	Y	Y	Y	H	Yes
Panda et al. (2021) [80]	Flex sensors	SVM/KNN/ANN	Y	Y	Y	N	M	Yes
Lee et al. (2022) [16]	5 resistive flex	SVM	Y	Y	Y	N	M	Yes
Ahmed et al. (2021) [66]	Flex + servo motors	Mirror rehab. System	Y	Y	Y	N	M	Yes
Yudhana et al. (2018) [98]	IMU + 5x flex	KNN classifier	Y	Y	Y	N	M	Yes
Lygouras (2003) [105]	Flex + strain gauge	Force measurement	Y	N	Y	N	M	Yes
Dhiman et al. (2025) [89]	IoT + ESP32 + sensors	Real-time monitoring	Y	Y	Y	N	M	Yes
Hosie et al. (2025) [90]	5 flex + IMU	Calibration-based	Y	Y	Y	Y	H	Yes
Imtiaz et al. (2023) [88]	Flex + FSR + ESP32	Multi-function glove	Y	Y	Y	N	M	Yes
Fang et al. (2025) [104]	Flex + inertial sensors	Full-body motion track	Y	Y	Y	Y	H	Yes
Flynn et al. (2013) [76]	Textile sensors	Signal processing	Y	N	Y	N	M	Yes
Plawiak et al. (2016) [73]	Flex sensors	SVM / neural network	Y	Y	Y	Y	H	Yes
Fujiwara et al. (2013) [75]	IMU sensors	Gesture classification.	Y	Y	Y	N	M	Yes
Kortier et al. (2014) [77]	IMU + flex	Finger motion capture	Y	Y	Y	Y	H	Yes
Kanawaday & Sane (2017) [78]	Flex sensors	ANN classifier	Y	Y	Y	N	M	Yes
Parvini et al. (2009) [79]	22 flex sensors.	MLP + GRUBC	Y	Y	Y	N	M	Yes
Islam et al. (2025) [85]	Flex + ESP32	Edge AI monitoring	Y	Y	Y	N	M	Yes
Bhaskaran et al. (2022) [103]	Flex + IMU	Deep learning	Y	Y	Y	N	M	Yes

Criteria: C1 = Clarity of sensor configuration; C2 = Description of AI/ML methodology; C3 = Reporting of experimental validation; C4 = Generalizability of findings. Rating: H = High (3–4 criteria met); M = Moderate (2 criteria met); L = Low (0–1 criterion met). Comparative synthesis was performed for only studies rated M or H

✓ = meets criterion; ✗ = does not meet criterion. Four criteria: C1 = Clarity of sensor configuration, C2 = Description of AI/ML methodology, C3 = Reporting experimental validation, and C4 = Generalizability. High (H): 3–4 criteria met; moderate (M): 2 criteria met. Twenty-four studies were shown, and all passed the quality threshold and were included in the comparative synthesis

Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
SVM	Support Vector Machine
CNN	Convolutional Neural Network
FSR	Force Sensitive Resistor
LSTM	Long Short -Term Memory
IMU	Inertial Measurement Unit
EMG	Electromyography
MCU	Microcontroller Unit
ESP32	Wi-Fi and Bluetooth Enabled Microcontroller
IoT	Internet of Things
ACC	Accelerometer
GYRO	Gyroscope
RF	Random Forest
KNN	K-Nearest Neighbors
ROM	Range of Motion
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
IC	Integrated Circuit
HCI	Human-Computer Interaction
ACC	Accuracy
OLED	Organic Light Emitting Diode
GRU	Gated Recurrent Unit
TP4050	Li-ion Charging Module
Li-Po	Lithium Polymer Battery
FMA	Fugl-Meyer Assessment
PPG	Photoplethysmography

Supplementary Information

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Supplementary Material 1.

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AM conducted the literature review, data collection, analysis, and manuscript drafting. AMA contributed to the study conception, critical revision, and final approval of the manuscript. All authors read and approved the final manuscript.

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Data availability
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Declarations

Competing interests
The authors have no conflict of interest to declare.

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References

1. Lin B-S, Lee IJ, Chiang P-Y, Huang S-Y, Peng C-W (2019) A Modular Data Glove System for Finger and Hand Motion Capture Based on Inertial Sensors. J Med Biol Eng 39(4):532–540. https://doi.org/10.1007/s40846-018-0434-6
2. Salawu FK, Danburam A, Olokoba AB (2010) Non-Motor Symptoms of Parkinson's Disease: Diagnosis and Management. Niger J Med 19(2):13. https://doi.org/10.4314/njm.v19i2.56496
3. Oess NP, Wanek J, Hedel HJAv (2010) Enhancement of bend sensor properties as applied in a glove for use in neurorehabilitation settings, in 2010 Annual International Conference of the IEEE Engineering in Medicine and Biology, 31 Aug.-4 Sept 2010. p 5903–5906. https://doi.org/10.1109/IEMBS.2010.5627534
4. Chanu T (2017) Comparative Analysis of Vision-Based and Smart Glove-Based Systems for Gesture Recognition. Int J Adv Res Comput Sci Softw Eng 7
5. Brigante CMN, Abbate N, Basile A, Faulisi AC, Sessa S (2011) Towards Miniaturization of a MEMS-Based Wearable Motion Capture System. IEEE Trans Industr Electron 58(8):3234–3241. https://doi.org/10.1109/TIE.2011.2148671

6. Zhang X, Chen X, Li Y, Lantz V, Wang K, Yang J (2011) A framework for hand gesture recognition based on accelerometer and EMG sensors. IEEE Trans Syst Man Cybernetics-Part A: Syst Hum 41(6):1064–1076
7. ASL et al (2022) Machine Learning-Based Smart Glove Using Flex Sensors for Gesture Detection. Sensors 22
8. Davarzani S, Pajouh MAA (2020) Design and fabrication of sensing system for rehabilitation of finger, in (2020) 28th Iranian Conference on Electrical Engineering (ICEE) IEEE, p 1–5
9. Wang Q, Liu Y (2018) Review of optical fiber bending/curvature sensor. Measurement 130:161–176. https://doi.org/10.1016/j.measurement.2018.07.068
10. Kumar S, Sultan MJ, Ullah A, Zameer S, Siddiqui S, Sami SK (2018) Human Machine Interface Glove Using Piezoresistive Textile Based Sensors. IOP Confer Ser Mater Sci Eng 414(1):012041. https://doi.org/10.1088/1757-899X/414/1/012041
11. Dipietro L, Sabatini AM, Dario P (2003) Evaluation of an instrumented glove for hand-movement acquisition. J Rehabilitation Res Dev 40:2
12. Williams N, Penrose J, Caddy C, Barnes E, Hose D, Harley P (2000) A goniometric glove for clinical hand assessment: construction, calibration and validation. J Hand Surgery: Br Eur Volume 25(2):200–207
13. Williams N (1997) The virtual hand: The Pulvertaft prize essay for 1996. J Hand Surg 22(5):560–567
14. Pasquale G, Mastrototaro L (2018) Glove-based systems for medical applications: Review of recent advancements. J Text Eng Fashion Technol 4(3):253–262
15. Pandey SK, Bhan A, Menon AK, Singh C (2020) Smart gloves with health monitoring and security. Int Res J Eng Technol 7(5):5661–5665
16. Lee KT, Chee PS, Lim EH, Lim CC (2022) Artificial intelligence (AI)-driven smart glove for object recognition application. Mater Today: Proc 64:1563–1568
17. Urav (2024) Deep Learning-Enabled Smart Glove for Real-Time Sign Language Translation. J Electr Syst 20
18. Alshami A, Nashwan A, AlDardour A, Qusini A (2025) Artificial Intelligence in Rehabilitation: A Narrative Review on Advancing Patient Care. Rehabilitación 59(2):100911. https://doi.org/10.1016/j.rh.2025.100911
19. Takahashi CD, Der-Yeghiaian L, Le V, Motiwala RR, Cramer SC (2008) Robot-based hand motor therapy after stroke. Brain 131(2):425–437
20. Houwink A, Nijland RH, Geurts AC, Kwakkel G (2013) Functional recovery of the paretic upper limb after stroke: who regains hand capacity? Arch Phys Med Rehabil 94(5):839–844
21. Prevention C (2007) Outpatient rehabilitation among stroke survivors: 21 states and the District of Columbia. MMWR Morb Mortal Wkly Rep no. 56:504–507
22. Coupar F, Pollock A, Rowe P, Weir C, Langhorne P (2012) Predictors of upper limb recovery after stroke: a systematic review and meta-analysis. Clin Rehabil 26(4):291–313
23. Chen X et al (2018) Therapeutic effects of sensory input training on motor function rehabilitation after stroke. Medicine 97(48):e13387. https://doi.org/10.1097/md.0000000000013387
24. Li S (2017) Spasticity, motor recovery, and neural plasticity after stroke. Front Neurol 8:120
25. Zorowitz RD, Gillard PJ, Brainin M (2013) Poststroke spasticity: sequelae and burden on stroke survivors and caregivers. Neurology 80(3supplement2):S45–S52
26. Feldman R, Young R, Koella W (1980) Spasticity: Disordered Motor Control Symposia Specialists Inc, Miami
27. Langhorne P, Coupar F, Pollock A (2009) Motor recovery after stroke: a systematic review. Lancet Neurol 8(8):741–754
28. Worsnopp T, Peshkin M, Colgate J, Kamper D (2007) An actuated finger exoskeleton for hand rehabilitation following stroke, in (2007) IEEE 10th international conference on rehabilitation robotics, IEEE, p 896–901 https://doi.org/10.1109/ICORR.2007.4428530
29. Kabir R, Sunny MSH, Ahmed HU, Rahman MH (2022) Hand rehabilitation devices: a comprehensive systematic review. Micromachines 13(7):1033
30. Laskowitz D, Grant G (2016) Translational Res Trauma brain injury
31. Nahmani M, Turrigiano GG (2014) Adult cortical plasticity following injury: Recapitulation of critical period mechanisms? Neuroscience 283:4–16. https://doi.org/10.1016/j.neuroscience.2014.04.029
32. Carmichael ST (2003) Plasticity of cortical projections after stroke. Neuroscientist 9(1):64–75
33. Keci A, Tani K, Xhema J (2019) Role of rehabilitation in neural plasticity. Open access Macedonian J Med Sci 7(9):1540
34. Kleim JA (2011) Neural plasticity and neurorehabilitation: Teaching the new brain old tricks. J Commun Disord 44(5):521–528 2011/09/01/. https://doi.org/10.1016/j.jcomdis.2011.04.006
35. Landry MD et al (2016) 2010–2020: Annual Supply and Demand Forecasting Models for Physical Therapists Across the United States. Phys Ther 96(1):71–80. https://doi.org/10.2522/ptj.20150010
36. Huang Y-C, Chen P-C, Tso H-H, Yang Y-C, Ho T-L, Leong C-P (2019) Effects of kinesio taping on hemiplegic hand in patients with upper limb post-stroke spasticity: a randomized controlled pilot study. Eur J Phys Rehabil Med 55(5):551–557
37. Villafañe JH et al (2018) Efficacy of short-term robot-assisted rehabilitation in patients with hand paralysis after stroke: a randomized clinical trial. Hand 13(1):95–102
38. Molteni F, Gasperini G, Cannaviello G, Guanziroli E (2018) Exoskeleton and end-effector robots for upper and lower limbs rehabilitation: narrative review. PM&R 10(9):S174–S188
39. Gull M, Bai S, Bak T (2020) A review on design of upper limb exoskeletons. Robotics 9(1):16
40. Sandison M et al (2020) HandMATE: wearable robotic hand exoskeleton and integrated android app for at-home stroke rehabilitation, in (2020) 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), IEEE, p 4867–4872 https://doi.org/10.1109/EMBC44109.2020.9175332
41. Huang X, Naghdy F, Naghdy G, Du H, Todd C (2018) The combined effects of adaptive control and virtual reality on robot-assisted fine hand motion rehabilitation in chronic stroke patients: a case study. J Stroke Cerebrovasc Dis 27(1):221–228
42. Borboni A, Mor M, Faglia R (2016) Gloreha—hand robotic rehabilitation: Design, mechanical model, and experiments. J Dyn Syst Meas Contr 138(11):111003
43. Fischer HC et al (2015) Use of a portable assistive glove to facilitate rehabilitation in stroke survivors with severe hand impairment. IEEE Trans Neural Syst Rehabil Eng 24(3):344–351
44. Schabowsky CN, Godfrey SB, Holley RJ, Lum PS (2010) Development and pilot testing of HEXORR: hand EXOskeleton rehabilitation robot. J Neuroeng Rehabil 7(1):36
45. Forducey P et al (2006) Tele-rehabilitation using the Rutgers Master II glove following carpal tunnel release surgery, in 2006 International Workshop on Virtual Rehabilitation, IEEE, p 88–93 https://doi.org/10.1109/IWVR.2006.1707533

46. Yap HK, Lim JH, Nasrallah F, Yeow C-H (2017) Design and preliminary feasibility study of a soft robotic glove for hand function assistance in stroke survivors. Front NeuroSci 11:547
47. Nilsson M, Ingvast J, Wikander J, von Holst H (2012) The Soft Extra Muscle system for improving the grasping capability in neurological rehabilitation, in 2012 IEEE-EMBS conference on biomedical engineering and sciences, IEEE, p 412–417
48. Haghshenas-Jaryani M, Nothnagle C, Patterson RM, Bugnariu N, Wijesundara MB (2017) Soft robotic rehabilitation exoskeleton (rehab glove) for hand therapy, in International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, vol 58158. American Society of Mechanical Engineers, p V003T13A005 https://doi.org/10.1115/DETC2017-68291
49. Radder B et al (2019) Home rehabilitation supported by a wearable soft-robotic device for improving hand function in older adults: A pilot randomized controlled trial. PLoS ONE 14(8):e0220544
50. Yurkewich A et al (2020) Myoelectric untethered robotic glove enhances hand function and performance on daily living tasks after stroke. J rehabilitation Assist Technol Eng 7:2055668320964050
51. Zhou Y, Desplenter T, Chinchalkar S, Trejos AL (2019) A wearable mechatronic glove for resistive hand therapy exercises, in (2019) IEEE 16th International Conference on Rehabilitation Robotics (ICORR), IEEE, p 1097–1102
52. In H, Kang BB, Sin M, Cho K-J (2015) Exo-glove: A wearable robot for the hand with a soft tendon routing system. IEEE Rob Autom Magazine 22(1):97–105
53. Hesse S, Kuhlmann H, Wilk J, Tomelleri C, Kirker SG (2008) A new electromechanical trainer for sensorimotor rehabilitation of paralyzed fingers: A case series in chronic and acute stroke patients. J Neuroeng Rehabil 5(1):21
54. Kuswanto D, Iskandriawan B, Mahardhika PS (2018) Power grip exoskeleton design as rehabilitation devices for post-stroke survivors, in (2018) 1st International Conference on Bioinformatics, Biotechnology, and Biomedical Engineering—Bioinformatics and Biomedical Engineering, vol 1. IEEE, p 1–6 https://doi.org/10.1109/BIOMIC.2018.8610556
55. Gasser BW, Bennett DA, Durrough CM, Goldfarb M (2017) Design and preliminary assessment of Vanderbilt hand exoskeleton, in (2017) International Conference on Rehabilitation Robotics (ICORR), IEEE, p 1537–1542 https://doi.org/10.1109/ICORR.2017.8009466
56. Dovat L et al (2008) HandCARE: a cable-actuated rehabilitation system to train hand function after stroke. IEEE Trans Neural Syst Rehabil Eng 16(6):582–591
57. Thielbar KO et al (2016) Benefits of using a voice- and EMG-driven actuated glove to support occupational therapy for stroke survivors. IEEE Trans Neural Syst Rehabil Eng 25(3):297–305
58. Wang D, Meng Q, Meng Q, Li X, Yu H (2018) Design and development of a portable exoskeleton for hand rehabilitation. IEEE Trans Neural Syst Rehabil Eng 26(12):2376–2386
59. Connelly L, Jia Y, Toro ML, Stoykov ME, Kenyon RV, Kamper DG (2010) A pneumatic glove and immersive virtual reality environment for hand rehabilitative training after stroke. IEEE Trans Neural Syst Rehabil Eng 18(5):551–559
60. Lu Z, Tong K-y, Shin H, Li S, Zhou P (2017) Advanced myoelectric control for robotic hand-assisted training: outcome from a stroke patient. Front Neurol 8:107
61. Lambercy O, Dovat L, Gassert R, Burdet E, Teo CL, Milner T (2007) A haptic knob for rehabilitation of hand function. IEEE Trans Neural Syst Rehabil Eng 15(3):356–366
62. Wege A, Zimmermann A (2007) Electromyography sensor-based control for a hand exoskeleton, in (2007) IEEE International Conference on Robotics and Biomimetics (ROBIO), IEEE, p 1470–1475
63. Sandoval-Gonzalez O et al (2016) Design and development of a hand exoskeleton robot for active and passive rehabilitation. Int J Adv Rob Syst 13(2):66
64. Meng Q, Shen Z, Nie Z, Meng Q, Wu Z, Yu H (2021) Modeling and evaluation of a novel hybrid-driven compliant hand exoskeleton based on a human-machine coupling model. Appl Sci 11(22):10825
65. Meeker C, Park S, Bishop L, Stein J, Ciocarlie M (2017) EMG pattern classification to control a hand orthosis for functional grasp assistance after stroke, in (2017) international conference on rehabilitation robotics (ICORR), IEEE, p 1203–1210
66. Ahmed T et al (2021) Flexohand: A hybrid exoskeleton-based novel hand rehabilitation device. Micromachines 12(11):1274
67. Delph MA, Fischer SA, Gauthier PW, Luna CHM, Clancy EA, Fischer GS (2013) A soft robotic exomusculature glove with integrated sEMG sensing for hand rehabilitation, in (2013) IEEE 13th International Conference on Rehabilitation Robotics (ICORR), IEEE, p 1–7 https://doi.org/10.1109/ICORR.2013.6650426
68. Dipietro L, Sabatini AM, Dario P (2008) A survey of glove-based systems and their applications. IEEE Trans Syst Man Cybernetics Part C (Applications Reviews) 38(4):461–482
69. Lee M, Bae J (2020) Deep learning based real-time recognition of dynamic finger gestures using a data glove. IEEE Access 8:219923–219933
70. Lin BS, Lee IJ, Chen JL (2020) Novel Assembled Sensorized Glove Platform for Comprehensive Hand Function Assessment by Using Inertial Sensors and Force Sensing Resistors. IEEE Sens J 20(6):3379–3389. https://doi.org/10.1109/JSEN.2019.2958533
71. Abbas S, Condell J, Gardiner P, McCann M, Todd S, Connolly J (2020) Can multiple wearable sensors be used to detect the early onset of Parkinson's Disease? in 2020 31st Irish Signals and Systems Conference (ISSC), IEEE, p 1–6 https://doi.org/10.1109/ISSC49989.2020.9180191
72. O'Flynn B, Torres J, Connolly J, Condell J, Curran K, Gardiner P (2013) Novel smart sensor glove for arthritis rehabilitation, in (2013) IEEE International Conference on Body Sensor Networks, IEEE, p 1–6
73. Pławiak P, Sośnicki T, Niedźwiecki M, Tabor Z, Rzecki K (2016) Hand Body Language Gesture Recognition Based on Signals From Specialized Glove and Machine Learning Algorithms. IEEE Trans Industr Inf 12(3):1104–1113. https://doi.org/10.1109/TII.2016.2550528
74. Edmison J, Jones M, Nakad Z, Martin T (2002) Using piezoelectric materials for wearable electronic textiles, in Proceedings. Sixth International Symposium on Wearable Computers, p 41–48. https://doi.org/10.1109/ISWC.2002.1167217
75. Fujiwara E, Miyatake DY, Santos MFMd, Suzuki CK (2013) Development of a glove-based optical fiber sensor for applications in human-robot interaction, in (2013) 8th ACM/IEEE International Conference on Human-Robot Interaction (HRI), p 123–124. https://doi.org/10.1109/HRI.2013.6483532
76. Flynn BO, Torres J, Connolly J, Condell J, Curran K, Gardiner P (2013) Novel smart sensor glove for arthritis rehabilitation, in (2013) IEEE International Conference on Body Sensor Networks, p 1–6. https://doi.org/10.1109/BSN.2013.6575529

77. Kortier HG, Sluiter VI, Roetenberg D, Veltink PH (2014) Assessment of hand kinematics using inertial and magnetic sensors. J Neuroeng Rehabil 11:70. https://doi.org/10.1186/1743-0003-11-70
78. Kanawaday A, Sane A (2017) Machine learning for predictive maintenance of industrial machines using IoT sensor data, in (2017) 8th IEEE International Conference on Software Engineering and Service Science (ICSESS), IEEE, p 87–90
79. Parvini F, McLeod D, Shahabi C, Navai B, Zali B, Ghandeharizadeh S (2009) An approach to glove-based gesture recognition, in International Conference on Human-Computer Interaction, Springer,p 236–245
80. Panda AK, Chakravarty R, Moulik S (2021) Hand gesture recognition using flex sensor and machine learning algorithms, in 2020 IEEE-EMBS Conference on Biomedical Engineering and Sciences (IECBES), IEEE, p 449–453 https://doi.org/10.1109/IECBES48179.2021.9398789
81. Faisal MAA, Abir FF, Ahmed MU, Ahad MAR (2022) Exploiting domain transformation and deep learning for hand gesture recognition using a low-cost dataglove. Sci Rep 12(1):21446
82. Ji A et al (2023) Dataglove for Sign Language Recognition of People with Hearing and Speech Impairment via Wearable Inertial Sensors. Sensors 23(15):6693. Available: https://www.mdpi.com/1424-8220/23/15/6693
83. Ravan M, Abbasnia A, Davarzani S, Amineh RK (2025) A Smart Glove Based on Inductive Sensors for Hand Gesture Recognition. IEEE Trans Human-Machine Syst, 55:549–558 https://doi.org/10.1109/THMS.2025.3566941.
84. Wang S et al (2024) Recognizing wearable upper-limb rehabilitation gestures by a hybrid multi-feature neural network. Eng Appl Artif Intell 127:107424. https://doi.org/10.1016/j.engappai.2023.107424
85. Islam I, See CH, Yu H (2025) Design and Validation of Data Glove with Flex Sensor and 9-Axis IMU to Assess Hand Dexterity, in (2025) International Conference on Software, Knowledge, Information Management & Applications (SKIMA), IEEE, p 1–5 https://doi.org/10.1109/SKIMA66621.2025.11155286
86. Ahmed W, Sattar MK, Shahnawaz W, Saeed U, Khan SM, Khan NA (2021) Wearable hand-rehabilitation system with soft gloves for patient with face paralysis and disability. Engin Proc 12(1):56
87. Ilyas SM, Jawed SF, Shakeel CS, Bawany LH, Amin R (2022) Design and development of a stroke rehabilitation glove for measuring and monitoring hand motions. Pakistan J Rehabilitation 11(2):177–188
88. Imtiaz (2023) Rehabilitation for Stroke Survivors: The Development of a Smart Glove, International Conference on Computer and Information Technology (ICCIT) https://doi.org/10.1109/ICCIT60459.2023.10441307
89. Dhiman H, Kumar R (2025) An Internet of Things-enabled smart glove for brain stroke rehabilitation. Measur Digit 1:100001. https://doi.org/10.1016/j.meadig.2025.100001
90. Hosie O, Isaksson M, McCormick J, Tirosh O, Hensman C (2025) The Design and Validation of an Open-Palm Data Glove for Precision Finger and Wrist Tracking. Sensors 25(2):367 Available: https://www.mdpi.com/1424-8220/25/2/367
91. Zheng R (2025) et al FIP: Endowing Robust Motion Capture on Daily Garment by Fusing Flex and Inertial Sensors, in Proceedings of the (2025) CHI Conference on Human Factors in Computing Systems, p 1–21 https://doi.org/10.48550/arXiv.2502.15058
92. Braun A, Wichert R, Kuijper A, Fellner DW (2014) A benchmarking model for sensors in smart environments, in European Conference on Ambient Intelligence, Springer, p 242–257
93. Fang B, Sun F, Liu H, Guo D (2015) A Novel Data Glove Design Based on Inertial and Magnetic Sensors. Int J Swarm Intell Evol Comput 4:1–2
94. Wang Q, Markopoulos P, Yu B, Chen W, Timmermans A (2017) Interactive wearable systems for upper body rehabilitation: a systematic review. J Neuroeng Rehabil 14(1):20. https://doi.org/10.1186/s12984-017-0229-y
95. Routhier F et al (2020) Clinicians' perspectives on inertial measurement units in clinical practice. PLoS ONE 15(11):e0241922. https://doi.org/10.1371/journal.pone.0241922
96. Tarchanidis KN, Lygouras JN (2003) Data glove with a force sensor. IEEE Trans Instrum Meas 52(3):984–989
97. Bhaskaran KA, Nair AG, Ram KD, Ananthanarayanan K, Vardhan HRN (2016) Smart gloves for hand gesture recognition: Sign language to speech conversion system, in (2016) International Conference on Robotics and Automation for Humanitarian Applications (RAHA), p 1–6. https://doi.org/10.1109/RAHA.2016.7931887
98. Yudhana A, Rahmawan J, Negara CUP (2018) Flex sensors and MPU6050 sensors responses on smart glove for sign language translation, in IOP conference series: materials science and engineering, vol 403, no. 1. IOP Publishing, p 012032 https://doi.org/10.1088/1757-899X/403/1/012032
99. Rajamohan A, Hemavathy R, Dhanalakshmi M (2013) Deaf-mute communication interpreter. Int J Sci Eng Technol 2(5):336–341
100. Chuang W-C, Hwang W-J, Tai T-M, Huang D-R, Jhang Y-J (2019) Continuous Finger Gesture Recognition Based on Flex Sensors, Sensors 19(18):3986. Available: https://www.mdpi.com/1424-8220/19/18/3986
101. Mehdi SA, Khan YN (2002) Sign language recognition using sensor gloves, in Proceedings of the 9th International Conference on Neural Information Processing, 2002. ICONIP'02., vol 5. IEEE, p 2204–2206 https://doi.org/10.1109/ICONIP.2002.1201884
102. N. T. Muralidharan, S. R. R, R.R. M, M. S. N, and E. H. M, "Modelling of Sign Language Smart Glove Basedon Bit Equivalent Implementation Using Flex Sensor," in *2022 International Conference on WirelessCommunications Signal Processing and Networking (WiSPNET)*, 24-26 March 20222022, pp. 99-104, doi: 10.1109/WiSPNET54241.2022.9767137.
103. Bhaskaran KA, Nair AG, Ram KD, Ananthanarayanan K, Vardhan HRN (2016) Smart gloves for hand gesture recognition: Sign language to speech conversion system. 2016 International Conference on Robotics and Automation for Humanitarian Applications (RAHA), pp. 1–6.https://doi.org/10.1109/RAHA.2016.7931887.
104. Zheng R, Fang J, Yao Y, Gao X, Zuo C, Guo S, Luo Y (2025) FIP: Endowing Robust Motion Capture on Daily Garment by Fusing Flex and Inertial Sensors. Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems. https://doi.org/10.1145/3706598.3714140.
105. Tarchanidis KN, Lygouras JN. Data glove with a force sensor. IEEE Transactions on Instrumentation and Measurement. 2003;52(3):984–989.

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